we always take $\log_2$

DSA - MISS UROOJ - Algorithm Analysis

Date ______ 20__

1. Analyze worst & best case for the algo: (Lecture #3 slides task)

// Compute real roots of quadratic eq.

$d \rightarrow \{(b*b) - (4*a*c)\}$ B = W = 4

if $d >= 0$ B = W = 2

B=0 W=2 $r[0] = (-b + sqrt(d))/(2*a)$

$r[1] = (-b - sqrt(d))/(2*a)$

return r B = W = 1

B.Case = 7 Worst Case = 21

2. Lecture #3.pdf (task)

Find asymptotic relation for T(n) in best & worst cases.

// subtracts an nxn matrix B from nxn matrix A.

// stores result in A.

Algo matSub (A,B,n).

n^2 for $i \leftarrow 0$ to $n-1$ do - n times.

for $j \leftarrow 0$ to $n-1$ do — n times

$A[i,j] \leftarrow A[i,j] - B[i,j]$

return.

T(n) for B & W : n^2.

3. Lecture #4.pdf (task)

Qi. Find the running time for data size n is increased 8 folds.

$n_0 = 8 \longrightarrow n = 8n$

a) $32\log n$

$= 32\log(8n) \Rightarrow 32\log(8+n)$

$= 32(\log 8 + \log n) \Rightarrow 32(3+\log n)$

$\boxed{= 96 + 32\log n}$

b) n $\boxed{8n}$

$\sqrt{n} \Rightarrow 2\sqrt{2n}$

c) $n + \log n$

$= 8n + \log(8n) \Rightarrow 8n + (\log 8 + \log n)$

$= 8n + (3 + \log n) \Rightarrow \boxed{8n + \log n + 3}$

d) $8n^2$

$= 8(8n)^2 \Rightarrow 8 \cdot 64n^2 \Rightarrow \boxed{512n^2}$

e) n^3

$= (8n)^3 = 512n^3$

f) $16n\log n$

$= 16(8n) \cdot \log(8n) \Rightarrow 128n \cdot (\log 8 + \log n)$

$= 128n(3 + \log n) \Rightarrow 384n + 128n\log n$

g) 4^n $\boxed{4^{8n}} \Rightarrow (4^8)^n = \boxed{65536^n}$

ii- How long will it take to execute on a 3.2GHz machine for the given i/p size. considering that operation takes 10 clock cycles

a) $T(n) = 5n^2 + 2n\sqrt{n}$, $n = 1000000$

- Calculate $T(n)$.

$T(1000000) = 5(1000000)^2 + 2(1000\cdots)$

$T(1000000) = 5.002 \times 10^{12}$

- Total clock cycles:

clock cycles = (cycle / operation) $\times$ $T(n)$

$= 10 \times T(n)$

$= 5.002 \times 10^{3}$

- Time in seconds:

$$\text{Time} = \frac{\text{clock cycle}}{\text{clockspeed}} = \frac{T(n)}{3.2 \times 10^9}$$

$\boxed{\text{Time} = 15631s}$

$\log_2$ = always

b) $T(n) = 3n\log n + 2n$ $n = 1000000$

$T(n) = 61794705.71$

clock cycle $= 10 \times T(n)$

$= 617947057.1$

$$\text{Time} = \frac{\text{clock cycle}}{\text{clockspeed.}} = \boxed{0.193s}$$

c) $T(n) = 2n - 1$, $n = 100$

$T(n) = 199$

clock cycle $= 10 \times T(n) = 1990$

$$\text{time} = \frac{\text{clock cycle}}{\text{clockspeed}} \quad \boxed{6.2 \times 10^{-7}s}$$

ii. How large a problem can it solve in 10 min. capable of executing basic operation in 5ns?

a. $T(n) = 0.5n^3 + 20$

- available time = 10 min or 600 sec
- in (ns) = $600 \times 10^9 = 6 \times 10^{11}s$
- No. of operations

$5ns \longrightarrow 1$ op.

$1ns \longrightarrow 1/5$ op

for $6 \times 10^{11} ns \longrightarrow 6 \times 10^{11}/5 = 1.2 \times 10^{11}$ op.

1.2×10^{11} operations can take place in $6 \times 10^{11}s$.

Now,

$0.5n^3 + 20 \leq 1.2 \times 10^{11}$

$0.5n^3 \leq 1.2 \times 10^{11} - 20$

$n^3 \leq 2.4 \times 10^{11}$

$n \leq \sqrt[3]{2.4 \times 10^{11}}$

$n \leq 6214$

b. $T(n) = 10\log n + 100$

$10\log n + 100 \leq 1.2 \times 10^{11}$

$10\log n' \leq 1.19 \times 10^{11}$

$= \log n \leq 1.19 \times 10^{11}$

$n \leq 10^{11/99,9,999,990}$

~~shouldn't we take~~ $\log_2$?

Omega → fn >
Big O → fn <

→ when dominant terms are changed little 'o' or little ω

DSA-MU- ASSIGNMENT #1

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LECTURE SLIDE #6 TASKS:

1. Provide values of c (or c_1 and c_2) and n_0 to prove.

a) $12n^3+5n^2+20n+30$ is $\Theta(n^3)$

Equal: $+30n$

$12n^3 \le 12n^3+5n^2+20n+30 \le 12n^3+5n^2+20n^2$

$\boxed{c_1 = 12 , c_2 = 67}$

$\frac{12n^3+5n^2+20n+30}{6n^3}$

$5n^2 \le 0$ $20n \le 0$ $= 36n^2+15n+20$

$n^2 \le 0$ $\boxed{n \le 0}$ $\frac{}{3n^2}$

$= 72n + 15$

$\boxed{n \le 0}$ $= \boxed{\frac{6n}{...} \in R}$

∵ n can never be zero, we will take $\boxed{n = 1}$

b) $3n^2+ n\log n$ is $\Omega(n^2)$

$\Rightarrow 3n^2 + n\log n \ge 3n^2$

$\because n\log n \le 0$: $n \le 0$, $\log n \le 0$

, $2^{\log n} \le 2^0$

$\boxed{c = 3 \text{ \& } n \le 1}$, $n \le 1$

c) $(n\log n)^2 + 0.01n^2$ is $O(n^3)$

$\because (n\log n)^2 \le n \le n^3$

$\therefore n\log n \le n^3$

$(n\log n)^2+0.01n^2 \le n^3+0.01n^2 \le n^3+0.01n^3$

$(n\log n)^2+ 0.01n^2 \le n^3+ 0.01n^3$

$(n\log n)^2+ 0.01n^2 \le (1.01)n^3$

$0.01n^2 \le 0$, $(n\log n)^2 \le 0$

$n^2 \le 0$ $n\log n \le 0$

$\boxed{n \le 0}$ $n \le 0$, $\log n \le 0$

$2^{\log n} \le 2^0$

$\boxed{n \le 1}$

$\boxed{c = 1.01 \quad n = 1}$

Q2. Using calculus, find assymptotic relations.

a) $f(n) = n^{1.5} + n\ln n$

$g(n) = n\log n$

Compare the dominant terms

$f(n) = n^{1.5}$ $g(n) = n\log n$

∴ change terms

Find the limit of the ratio

$$\frac{f(n)}{g(n)} = \frac{n^{1.5}+ n\ln(n)}{n\log n}$$

$$\Rightarrow \frac{n^{1.5}}{n\log n} + \frac{\cancel{n}\cdot \ln n}{\cancel{n}\cdot\log n}$$

$$\Rightarrow \frac{\sqrt{n}}{\log n} + \frac{\ln n}{\log n}$$

Evaluate limit to ∞.

$$\lim_{n\to\infty}\left[\frac{\sqrt{n}}{\log n} + \frac{\ln n}{\log n}\right] \quad \because \log_a b = \frac{\ln b}{\ln a}$$

=

$\because \frac{d}{dx}\ln(x) = \frac{1}{x}$ convert log to ln

$$= \lim_{n\to\infty}\left[\sqrt{n} \div \frac{\ln(n)}{\ln 2} + \cancel{\ln n} \div \frac{\cancel{\ln n}}{\ln 2}\right]$$

$$= \lim_{n\to\infty}\left[\frac{\ln 2 \cdot \sqrt{n}}{\ln n} + \ln 2\right]\left[\frac{\infty}{\infty}\right]$$

= Applying L-Hopital:

$$= \lim_{n\to\infty}\left[\frac{\ln 2}{2\sqrt{n}} \div \frac{1}{n} + \ln 2\right]$$

$$= \lim_{n\to\infty}\left[\frac{n\ln 2}{2\sqrt{n}} + \ln 2\right]$$

$\log_{10} n = \frac{\ln n}{\ln 10}$ $\quad \cdot \frac{\ln(\text{ans})}{\ln(\text{base})}$

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$\lim_{n\to\infty}\left[\frac{\sqrt{n}\cdot \ln 2}{2} + \ln 2\right]$ $\quad ∵ \sqrt{\infty} = \infty$, $\frac{\infty}{2} = \infty$ & $\infty + \ln 2 = \infty$

$= \infty$

we know if ans $= \infty$,

$f(n)$ is $\omega(g(n))$

b) $f(n) = (n^3 + 4n + 10)^{100}$,

$g(n) = n^3$

We can approximate:

$f(n) \sim (n^3)^{100} = n^{300}$

∵ dominant terms are dif:

$= \frac{f(n)}{g(n)} \Rightarrow \frac{(n^3+4n+10)^{100}}{n}$

∵ We deal with dominant term

$\approx n^{300}/n$

$\approx n^{299}$

Appling limit:

$\lim_{n\to\infty} n^{299} = \infty$

c) $f(n) = 2^{2n}$ $\quad g(n) = 2^n$

∵ dominant terms dif:

$\frac{f(n)}{g(n)} = \frac{2^{2n}}{2^n} \Rightarrow 2^{2n-n} \Rightarrow 2^n$

Applying limit: $2^{\infty} = \infty$

$f(n)$ is $\omega(g(n))$

d) $f(n) = 2^{2n-1} - 1$ $\quad g(n) = 2^{2n}$

∵ dominant terms diff

$\frac{f(n)}{g(n)} = \frac{2^{2n-1} - 1}{2^{2n}}$

$= (2^{2n-1} - 1)\, 2^{-2n}$

$= 2^{2n-2n-1} - 2^{-2n}$

$= 2^{-1} - 2^{-2n} \Rightarrow \frac{1}{2} - \frac{1}{2^n}$

Applying limit:

$= \frac{1}{2} - \frac{1}{\infty} \Rightarrow \frac{1}{2} - 0 \Rightarrow \frac{1}{2}$ Ans

∵ ans $\in \mathbb{R} \Rightarrow f(n)\ \Theta(g(n))$

e) $f(n) = \log_{10} n$, $g(n) = \log n$

$= f(n) = \frac{\ln n}{\ln 10}$ $\quad g(n) = \frac{\ln n}{\ln 2}$

Comparing:

$= \frac{\ln n}{\ln 10} \div \frac{\ln n}{\ln 2} \Rightarrow \frac{\ln 2}{\ln 10} \in \mathbb{R}$

$f(n)$ is $\Theta\, g(n)$

Apply c with greater func

$\left[\begin{matrix}\log^2 n \\ (\log n)^2\end{matrix}\right]$ same $\log^2 n = \log \cdot \log n$ $\quad \log^2 n \neq \log \log n$

$(\log n)^2$ $(\sin t)^2$

$\sin^2 t$ Date______20___

MISS UROOJ ASSIGNMENT 01

1a. Algorithm A uses $10n\log n$ operations, while algorithm B uses n^2 operations. Determine the value of c and n_0 such that A is better than B for $n \geq n_0$.

A : $10n\log n$ B : n^2

$10n\log n$ grows more slowly than n^2

$\Rightarrow 10n\log n \leq cn^2$

$10\log n \leq cn$

$n \geq 0 \Rightarrow n \geq 1 \quad n \neq 0$

$\boxed{c = 10 \quad n_0 = 1}$

b. Repeat part (a) with algo A using $40n^2$ while algo B using $2n^3$

We know that

$40n^2 \leq cn^3 \cdot 2$

$40 \leq 2cn \qquad 2n \geq 0$

$2c = 40 \qquad n \geq 1$

$\boxed{c = 20}$

2. If $p(n)$ is a polynomial in n of degree of m, then what is the big O of $\log p(n)$?

$p(n) = n^m C_m + n^{m-1} C_{m-1} + n^{m-2} C_{m-2} +$

$p(n)$ is $O(n^m)$

$\log p(n)$ is $\log O(n^m)$

is $O(\log n^m)$

$O(m\log n)$

$\because m$ is constant

$\log P(n)$ is $O(\log n)$

3.a Consider

$f_1(n) = 3n^2\log n$

$f_2(n) = n\log^2 n$

$f(n) = f_1(n) + f_2(n)$

show that $f(n) = Of_1(n)$

$f_1(n) \doteq f_1(n^2)$

$f_2(n) = f_2(n)$

$fn = 3n^2\log n + n\log^2 n$

$= O(n^2\log n + (n))$

fn is $O(\max\{f_1, f_2\})$

fn is $O(n^2\log n)$

$\because f(n)$ is $O(f_1(n))$

b. $f(n) = 10n^2 + 6n + 2$. Show that $f(n) = \Theta(n^2)$

$10n^2 \leq 10n^2 + 6n + 2 \leq 18n^2$

Therefore $f(n) = \Theta(n^2)$

or. (Better)

$\frac{f(n)}{n^2} = \frac{10n^2+6n+2}{n^2} = \frac{20n+6}{2n} =$

$\Rightarrow \frac{20}{2} = 10 \in R$ therefore $\Theta(n^2)$

To prove $f(n) = \Theta(n^2)$

4. Find big O not:

i: $6n\log n = O(n\log n)$

ii: $2^{\log n} = 2^{\log_2 n} = n = O(n)$

iii: $1/n = n^{-1} = O(n^{-1})$

iv: $2n \cdot \log^2 n = O(n\log^2 n)$

v: $n^2\log n = O(n^2\log n)$

vi: $2^{100} = O(1)$

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vii. $n\log_4 n \Rightarrow \log_4 n = \frac{\log_2 n}{\log_2 4}$

$\log_4 n = \frac{1}{2} \cdot \log_2 n \Rightarrow \frac{1}{2} n \cdot \log n$

$\Rightarrow O(n\log n)$

viii: $4n^{3/2}$ $O(n^{3/2})$

ix: 2^n $O(2^n)$

x: $n^{0.01}$ $O(n^{0.01})$

xi: $\log \cdot \log n$. $O(\log \cdot \log n)$

xii: $\lceil \sqrt{n} \rceil = O(n^{0.5})$

xiii. $3n^{0.5} = O(n^{0.5})$

xiv. $n^3 = O(n^3)$

xv. $\log^2 n = O(\log^2 n)$

Q5. Best case? Worst case?
Find complexity. Analyze the result for big O.

a. Func. to find the largest & smallest value in an array of numbers.

```
def largestsmallest (A,n):
    Largest = A[0] -> O(1) B/W
    smallest = A[0] -> O(1) B/W
    for i in range (1, n).          ] n(B)
        if A[i] > largest:          ]
            largest = A[i] - O(B)
    or
        else,                       ] n(B)
            if A[i] < smallest,     ]
                smallest = A[i] - O(B)
print (largest)
print (smallest)
```

Best Case	Worst Case
largest/smallest	
~~at A[0]~~	~~A[-1] or not~~
First cond. always T.	First cond. always F.

O(n) is O(n) as the code will run for the same time in both worst & best cases.

6. Func. to the n^{th} term of fibonacci series. using array

```
def fibarray(n):
    F[0] = 1          0 -> n+1
    F[1] = 1          2, 3, 4, 5
    for i in range (2, n+1):        ]
        F[i] = F[i-1] + F[i-2]      ] n-2
    return F[n]
```

Best Case = Worst case = O(n)

6. Assymptotic relations between f(n) and g(n).

a. $f(n) = 8n-2$ $g(n) = n$

∴ same dominant terms:

$\Rightarrow$ $8n - 2 \le 8n \le n$

$\Rightarrow$ $f(n) \le g(n)$

$f(n)$ is $O(g(n))$

don't do this always limit's method

b. $\lim_{n\to\infty} \left(\frac{8n-2}{n} \right) \Rightarrow \lim_{n\to\infty} \left[8 - \frac{2}{n} \right]$

Applying limit.

$= 8 \in C$ $f(n)$ is $\Theta g(n)$

$\frac{n}{2} - 2 \quad (\frac{n-4}{2})($

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b. $\log(n+1) = f(n) \quad g(n) = \log n$

Changing log:

$f(n) = \frac{\ln(n+1)}{\ln 2} \quad g(n) = \frac{\ln n}{\ln 2}$

$\lim_{n\to\infty}\left[\frac{\ln(n+1)}{\cancel{\ln 2}} \div \frac{\ln n}{\cancel{\ln 2}}\right]$

$\Rightarrow \lim_{n\to\infty}\left[\frac{\ln(n+1)}{\ln n}\right]$ Applying L-Hopital

$\Rightarrow \left[\frac{n}{(n+1)}\right] \Rightarrow \frac{1}{1} \in R$

$f(n)$ is $\Theta g(n)$

c. $f(n) = 9n^2 \quad g(n) = 4n^2+7$

$\lim_{n\to\infty}\left[\frac{9n^2}{4n^2+7}\right] \Rightarrow \lim_{n\to\infty}\left[\frac{18n}{8n+7}\right]$

$\Rightarrow \lim_{n\to\infty}\left[\frac{18}{8}\right] \in R$ $f(n)$ is $\Theta g(n)$

d. $f(n) = 10n$ and $g(n) = n^2 - 10n$

$\lim_{n\to\infty}\left[\frac{10n}{n^2-10n}\right] \Rightarrow \lim_{n\to\infty}\left[\frac{10}{n-10}\right]$

$\Rightarrow \frac{10}{\infty} \Rightarrow 0$ $f(n)$ is $o(g(n))$

e. $f(n) = n \quad g(n) = mn \quad m = c$

$\lim_{n\to\infty}\left[\frac{\cancel{n}}{m\cancel{n}}\right] \in R$ $f(n)$ is $\Theta(g(n))$

f. $f(n) = n \quad g(n) = n^n$

$\lim_{n\to\infty}\left[\frac{n}{n^n}\right] \Rightarrow \lim_{n\to\infty}\left[\frac{\log n}{n \cdot \log n}\right]$

$\lim_{n\to\infty}\left[\frac{1}{n}\right] \Rightarrow \frac{1}{\infty} = 0$ $f(n)$ is $o(g(n))$

g. $f(n) = n^3 \quad g(n) = n^2 \log n$

$\lim_{n\to\infty}\left[\frac{n^3}{n^2\log n}\right] \Rightarrow \lim_{n\to\infty}\left[\frac{n}{\log n}\right]$

$\Rightarrow \lim_{n\to\infty}\left[n \div \frac{\ln n}{\ln 2}\right] \Rightarrow$

$\Rightarrow \lim_{n\to\infty}\left[\frac{\ln 2 . n}{\ln n}\right]$ L-Hopital

$\Rightarrow \lim_{n\to\infty}\left[\ln 2 \cdot n\right] \Rightarrow \infty$

$f(n)$ is $\omega(g(n))$ or

$f(n)$ is $\omega(n^2 \log n)$

h. $f(n) = 2^n \quad g(n) = n$

$\lim_{n\to\infty}\left[\frac{2^n}{n}\right] \Rightarrow \lim_{n\to\infty}\left[\frac{n \cdot \log 2}{\log n}\right]$

$\Rightarrow \lim_{n\to\infty}\left[\frac{n \cdot \ln 2}{\ln 2} \div \frac{\ln n}{\ln 2}\right]$

$\Rightarrow \lim_{n\to\infty}\left[\frac{\ln 2 \cdot n}{\ln n}\right]$ L-Hopital

$\Rightarrow \lim_{n\to\infty}\left[\ln 2 \cdot n\right] \Rightarrow \infty$

$f(n)$ is $\omega(g(n))$

i. Find big O-not:

1. dot in GCR soln.

8. first loop = n odd = 2+-

first i odd = 2^1 2 → odd = 2^2

n → odd = 2^n

second loop : 2^n

max $(2^n, n)$

DSA - LECTURE#14.pdf   Home Work   Date ______ 20__

Q1. Convert Infix to Postfix expression:

Expression : 8+9-(5-2)×((1-7)+4)/2   20

89+ - 52-

Token	Stack	Result
8		8
+	+	8
9	+	89
−	−	89+ (+ = −)
(	−(	89+
5	−(	89+5
−	−(−	89+5
2	−(−	89+52
)	−	89+52−
×	−×	89+52− ∵ − < x
(	−×(	89+52−
(	−×((	89+52−
1	−×((	89+52−1
−	−×((−	89+52−1
7	−×((−	89+52−17
)	−×(	89+52−17−
+	−×(+	89+52−17−
4	−×(+	89+52−17−4 4
)	−×	89+52−17−4+
/	−/	89+52−17−4+× + × x = /
2	−/	89+52−17−4+2

Same precedence: pop.

/ −

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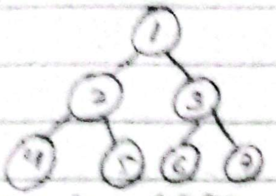
DSA pdf 19

Q1. Draw all possible ACBT's for 7, 11 & 13 nodes.

For 7 nodes:

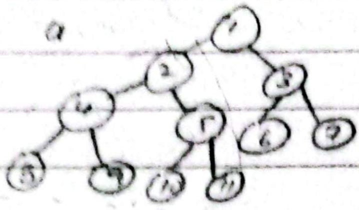
a.



PBT is ACBT

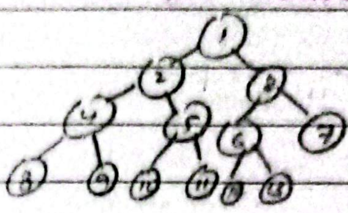
For 11 nodes:

a



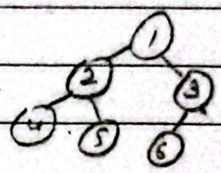
3 - 2 = 1

For 13 nodes:

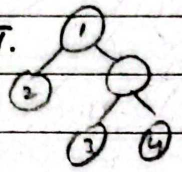


3 - 3 = 0 but left = PBT

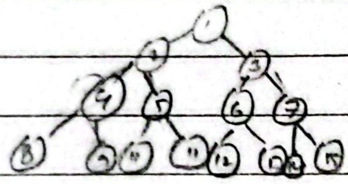
Q2. Draw ACBT which is not SBT.



Q3. Draw an SBT which is not ACBT.



Q4. Draw an PBT with 16<n<65



only two possible PBTs

31 nodes & 65

Q5. Tree with :

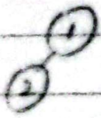
0 nodes.

PBT , SBT , ACBT
PBT

1 node.

PBT , SBT , ACBT

Q6. Draw an ACBT and SBT of 2 nodes.



ACBT & this can't be SBT

DSA pdf 21.

Q: Make BT of height 4 with right most tree node absent check if almost complete binary tree.

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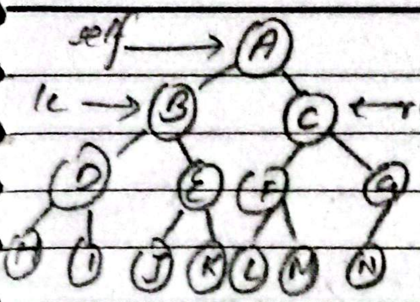
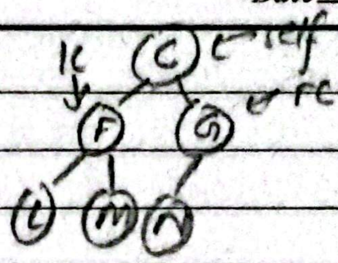
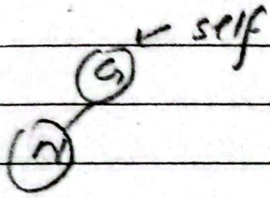


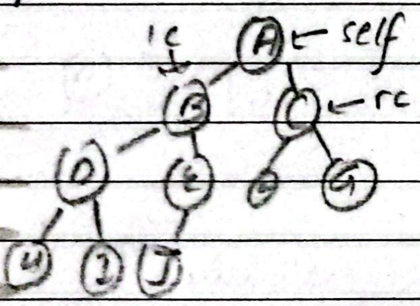
$hl = 3$
$hr = 3$
$hl = hr$ & isperfect() →



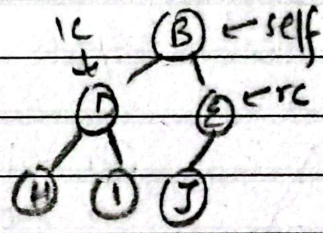
$lh = 2$ $hr = 2$
$lh = hr$ perfect



part (b) height = 4 & ⇒

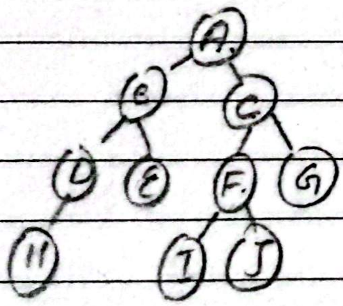


$hl = 3$
$hr = 2$ ⇒
rc → perfect ✓
$3 = 2+1$



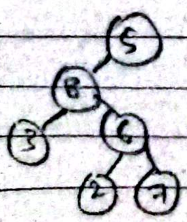
$hl = 2$ lc perfect
$hr = 2$

Q: Traverse for all three traversal orders



Inorder: HDBEAIFJCGA (LNR)
Preorder: ABDHECFIJG (NLR)
Post order: HDEBIJFGCA (LRN)

Q: Buildtree for val = [5, 8, None, 3, 6, None, None, 2, 7]



Date ______ 20 ___

Q: Run the search func. for the tree for target: B, F and J

(i) For B:

node = A

target = B

returned [node, 2]

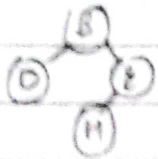
(ii) For F:

node : A

x = search(B, F) ii-a

search(C, F) ii-c x = [None, 0]

ii-a



x = search(D, F) ii-b

x = [None, 0]

search(E, F)

ii-b D

x = search (None, F)

x = [None, [0]

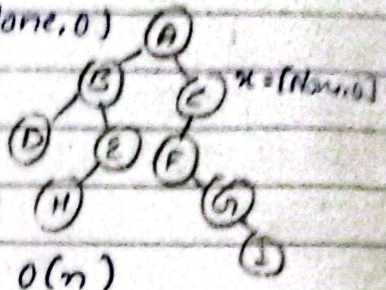
search (None, F)

ii-c return [Node, 1]

Final = [Node, 1]

iii. For J:

x [None, 0]



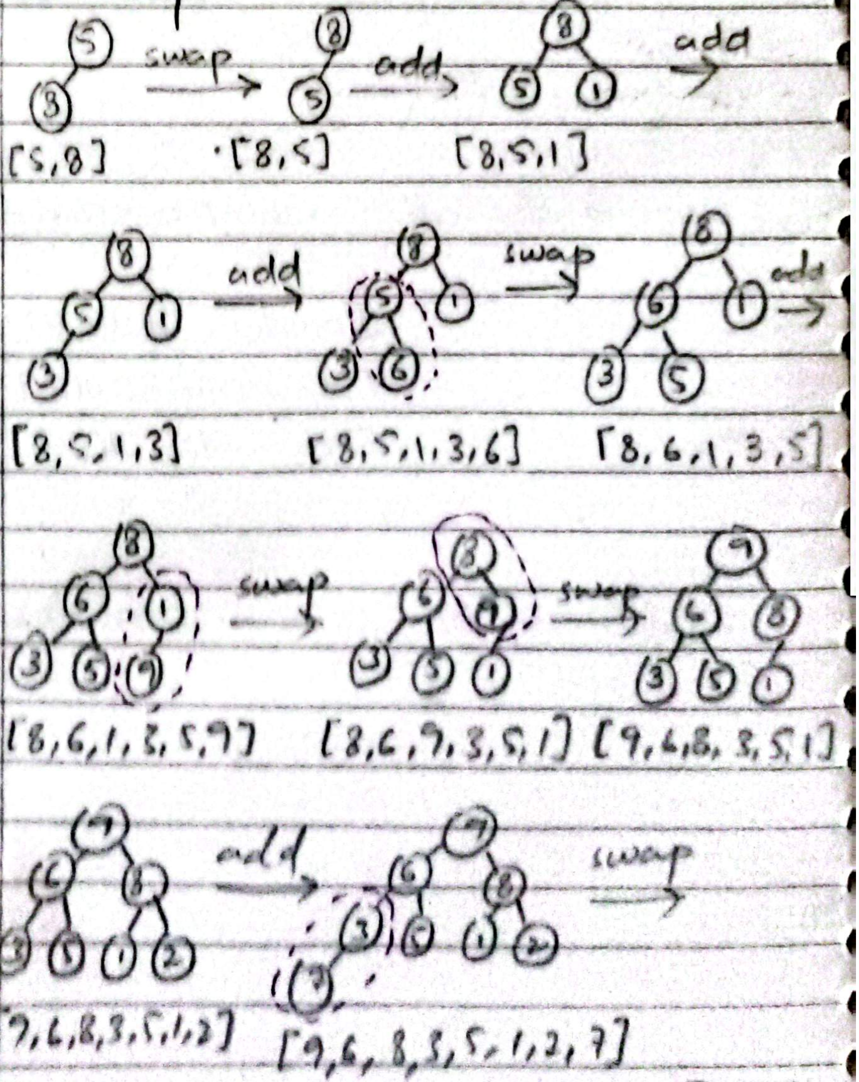
Worst Case:

Node data (target) not in tree. O(n)

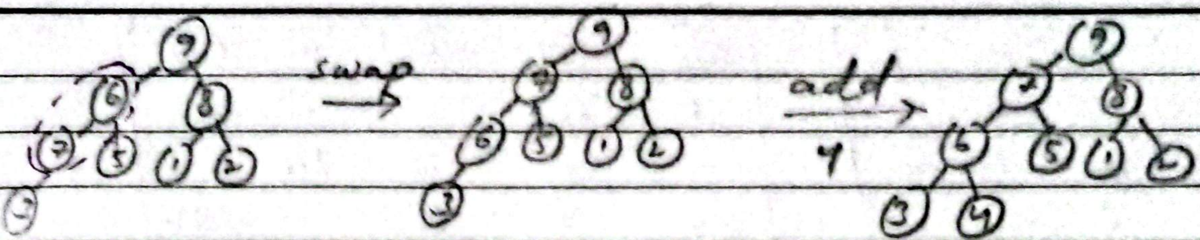
got to check all nodes.

Space Complexity: O(h)

Q: val = [5, 8, 1, 3, 6, 9, 2, 7, 4]. Build a heap.



Date______ 20___

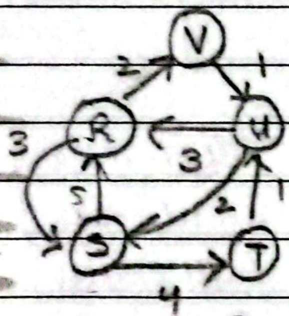


That's how it will be stored in memory

[9, 6, 8, 7, 5, 1, 2, 3] [9, 7, 8, 6, 5, 1, 2, 3] [9, 7, 8, 6, 5, 1, 2, 3, 4]

Lecture # 25.pdf

Q: For the weighted graph shown here, draw the weighted matrix & execute the shortest path algo.



W =

	R	S	T	U	V
R	0	3	0	0	2
S	5	0	4	0	0
T	0	0	0	1	0
U	3	2	0	0	0
V	0	0	0	1	0

– only direct paths are mentioned.

Q_0 =

	R	S	T	U	V
R	∞	3	∞	∞	2
S	5	∞	4	∞	∞
T	∞	∞	∞	1	∞
U	3	2	∞	∞	∞
V	∞	∞	∞	1	∞

–	RS	–	–	RV
SR	–	ST	–	–
–	–	–	TU	–
UR	US	–	–	–
–	–	–	VU	–

Q_1 =

	R	S	T	U	V
R	∞	3	∞	∞	2
S	5	8	4	∞	7
T	∞	∞	∞	1	∞
U	3	2	∞	∞	5
V	∞	∞	∞	1	∞
		8	∞	∞	7

–	RS	–	–	RV
SR	SRS	ST	–	SRV
–	–	–	TU	–
UR	US	–	[illegible]	URV
–	–	–	VU	–

Date ______ 20___

$$Q_2 = \begin{array}{c} \\ R \\ S \\ T \\ U \\ V \end{array}\begin{array}{c} \begin{array}{ccccc} R & S & T & U & V \end{array} \\ \left[\begin{array}{ccccc} 8 & 3 & 7 & \infty & 2 \\ 5 & 8 & 4 & \infty & 7 \\ \infty & \infty & \infty & 1 & \infty \\ 3 & 2 & 6 & \infty & 5 \\ \infty & \infty & \infty & 1 & \infty \end{array}\right] \end{array} \quad \left[\begin{array}{ccccc} RSR\ \cancel{SRS} & RS & RST & - & RV \\ SR & SRS & ST & - & SRV \\ - & - & - & TU & - \\ UR & US & UST & - & URV \\ - & - & - & VU & - \end{array}\right]$$

$$Q_3 = \begin{array}{c} \\ R \\ S \\ T \\ U \\ V \end{array}\begin{array}{c} \begin{array}{ccccc} R & S & T & U & V \end{array} \\ \left[\begin{array}{ccccc} 8 & 3 & 7 & 8 & 2 \\ 5 & 8 & 4 & 5 & 7 \\ \infty & \infty & \infty & 1 & \infty \\ 3 & 2 & 6 & 7 & 5 \\ \infty & \infty & \infty & 1 & \infty \end{array}\right] \end{array} \quad \left[\begin{array}{ccccc} RSR & RS & RST & RTU & RV \\ SR & SRS & ST & STU & SRV \\ - & - & - & TU & - \\ UR & US & UST & UTU & URV \\ - & - & - & VU & - \end{array}\right]$$

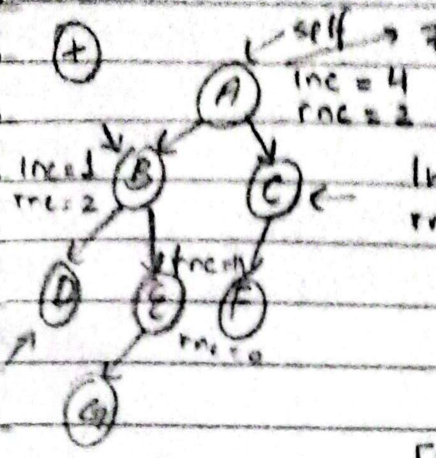
$$Q_4 = \begin{array}{c} \\ R \\ S \\ T \\ U \\ V \end{array}\begin{array}{c} \begin{array}{ccccc} R & S & T & U & V \end{array} \\ \left[\begin{array}{ccccc} 8 & 3 & 7 & 8 & 2 \\ 5 & 7 & 4 & 5 & 7 \\ 4 & 3 & 7 & 1 & 6 \\ 3 & 2 & 6 & 7 & 5 \\ 4 & 3 & 7 & 1 & 6 \end{array}\right] \end{array} \quad \left[\begin{array}{ccccc} RSR & RS & RST & RTU & RV \\ SR & SUS & ST & STU & SRV \\ TUR & TUS & TUT & TU & TUV \\ UR & US & UST & UTU & URV \\ VUR & VUS & VUT & VU & VUV \end{array}\right]$$

$$Q_5 = \begin{array}{c} \\ R \\ S \\ T \\ U \\ V \end{array}\begin{array}{c} \begin{array}{ccccc} R & S & T & U & V \end{array} \\ \left[\begin{array}{ccccc} 6 & 3 & 7 & 3 & 2 \\ 5 & 7 & 4 & 5 & 7 \\ 4 & 3 & 7 & 1 & 6 \\ 3 & 2 & 6 & 6 & 5 \\ 4 & 3 & 7 & 1 & 6 \end{array}\right] \end{array} \quad \left[\begin{array}{ccccc} RVR & RS & RST & RVU & RV \\ SR & SUS & ST & STU & SRV \\ TUR & TUS & TUT & TU & TUV \\ UR & US & UST & UVU & URV \\ VUR & VUS & VUT & VU & VUV \end{array}\right]$$

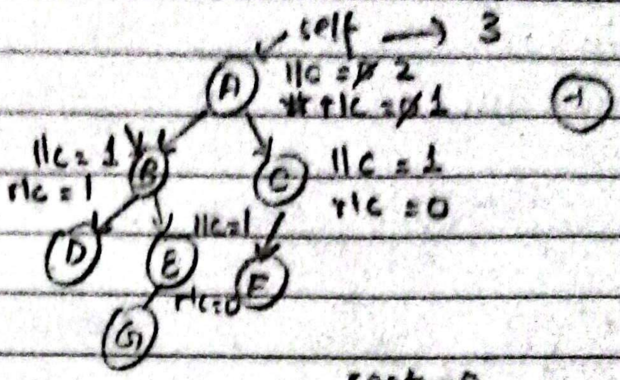
Date ______ 20__

DSA pdf: 20 Node Count

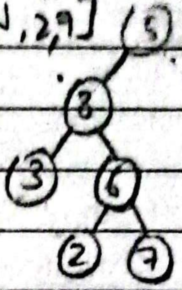
Q1. Run node count algo for:



Q2. Run code for leaf count algo



[5, 8, None, 3, 6, N, N, 2, 7]

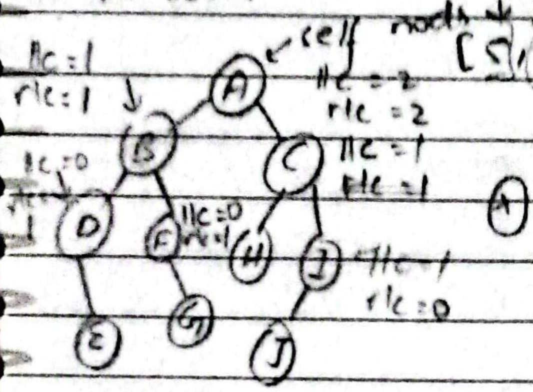


root = 5
nodes = [5]
n = 0
i = 1

[5, 8, N, 3, 6, N, N, 2, 7]
[5, 8, 3, 6]

leaf Count :

nodes [5, 8, 3, 6

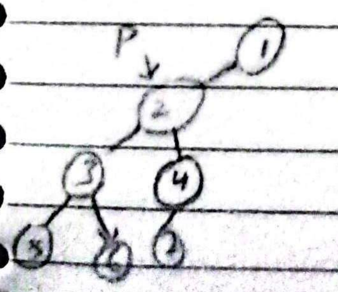


[1, 2, None, 3, 4, 5, 6, 7]

val = [5, 8, None, 3, 6, None, None, 2, 7]

root → 5

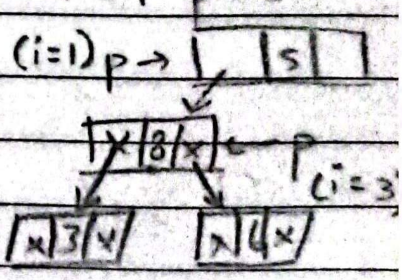
root = 5



n = 0 1 2
i = 1 2 3 4
nodes = [1]
[1, 2]
1, 2, 3, 4

nodes = [5] n = 0 i = 1 p = 5

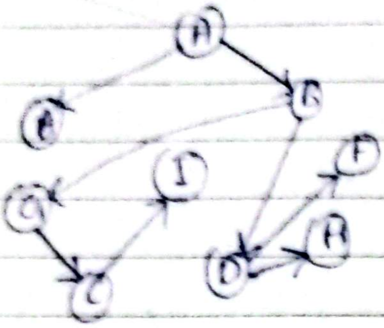
nodes =



nodes [5, 8]
i = 2
i = 3
n = 1
i = 4
i = 5
n = 2
nodes [5, 8, 3, 6]

Batch 2022 - Final Paper

Q6(a):
Spanning Tree



Q All nodes reacheable from C (6b)

A	B E I	1
B	D G	~~1~~ ~~2~~ 3 1: ~~C~~ ~~D~~ ~~I~~ ~~F~~ ~~H~~ ~~B~~ ~~G~~ E
C	D I	~~1~~ 3
D	F H	~~1~~ ~~2~~ 3
E	-	1
F	B H	~~1~~ ~~2~~ 3
G	C E I	1
H	-	~~1~~ ~~2~~ 3 C I D H F B G E
I	D	~~1~~ ~~2~~ 3

Q6c Apply selection sort on CAMPFIRE.

2 1 6 7 4 5 3

m = 7

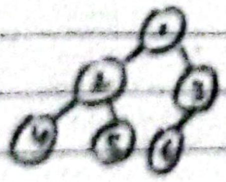
[2, 1, 6, 7, 4, 5, 3]
0 1 2 3 4 5 6

i = 0
j = 0
min pos: 0 1 [1, 2, 6, 7, 4, 5, 3]
1 2 3 (3) 4 5 6
i: ~~1~~ ~~2~~ 3
min_pos: ~~1~~ ~~2~~ ~~4~~ 6

m = len(values)
for i in range([illegible])
~~for j in 10~~
min pos = i
for j in range(i+1, m)
if values[min pos] > values[j]
min-pos = j
if [illegible] != j

Q5.a with 6 nodes draw an ACBT & CBT

ACBT



Can't CBT

Q5b Build a heap. HIJACKER
4 5 6 1 2 7 3 8